METHOD 11 / A COMPLETE BEGINNER GUIDE

Regime-conditioned correlation

Does correlation look different in calm weeks versus stress weeks? Regime-conditioned correlation computes Pearson-style correlation separately within states you define in advance.

You will see rho_quiet and rho_stress, why you must not average correlations, and how Benediktsdottir and Scotti (2009) motivate state-dependent FX co-movement. Teaching data only.

Read it in three parts

First: Idea, percents, data. **Next:** Research, symbols, formulas, hand calc. **Last:** Practice, limits, software.

Pages	What you will learn
2	Idea; is rho a percent?
3	FX percent example
4	Which series and defining states
5	Benediktsdottir & Scotti 2009
6-9	Symbols, formulas, hand calc
10-11	Use on majors and limits
12-14	Checks, software, sources

Prepared for the folder: 11 - Regime-Conditioned Correlation
3 November 2026

02 / THE BASIC IDEA

What does regime correlation do?

Split your sample into non-overlapping states (quiet, stress, or custom labels). Within each state, compute correlation of return series x and y on rows that belong to that state only.

Two scores, not one blended guess

```
rho_quiet = Corr(x, y | state = quiet)
rho_stress = Corr(x, y | state = stress)
```

Is this a percent?

No. Returns inside each state may be written as percents. rho_quiet and rho_stress are correlations on [-1, +1].

03 / RETURNS IN PLAIN NUMBERS

FX percent math on one move

```
Simple: 1.1110/1.1000 - 1 = 0.01 (1.00%)
Log: ln(1.1110/1.1000) = about 0.009950 (~0.995%)
```

Regime correlation uses return series. States are usually defined from vol indices, policy dates, or rules known without peeking at future pair-specific outcomes.

04 / DATA RULES

Which series and how to define states?

Returns, aligned times

Same bar close for both legs. One row per time index before you tag regimes.

Define states with known information

Examples: VIX above a threshold, realised vol quintile computed from past data only, or calendar windows around labelled events. Write the rule before you compute rho_stress.

Do not average correlations

The mean of rho_quiet and rho_stress is generally **not** equal to the full-sample correlation. Report each regime separately with its sample size n.

Research notes

Benediktsdottir, J., & Scotti, C. (2009) [1]

Working paper / IMF context on time-varying and state-dependent exchange rate correlations. They argue that pooling calm and crisis data hides economically different co-movement.

This folder uses their lesson at a simple level: conditional correlation by regime, not a full copy of their econometric model.

What we do not claim

We do not reproduce their full empirical tables here. Experiments in this folder have not started.

06 / DICTIONARY

Meet the symbols

Symbol	Meaning
S_t	Regime label at time t (quiet, stress, ...).
1{...}	Indicator: 1 if condition true, else 0.
rho_quiet	Correlation using rows with S_t = quiet only.
n_quiet	Count of paired returns in quiet regime.
x_t, y_t	Return series (same definition both legs).

The same symbols apply when you have more than two regimes; add rho and n for each label you use.

The main formulas

```
rho_quiet = Corr( {x_t, y_t} for t with S_t = quiet )
rho_stress = Corr( {x_t, y_t} for t with S_t = stress )
```

Corr here is the usual Pearson formula on the subsample. Spearman or robust variants are different methods (see folders 02 and 04).

Full sample vs regimes

```
Corr(x, y) on all rows != (rho_quiet + rho_stress) / 2 in general
```

Pool correlation weights every day once. Regime correlations answer different questions.

08 / DESIGN NOTES

Choosing regimes carefully

Overlap and gaps

If a day fits no state, drop it or add an other bucket. Document the rule.

Small n per regime

With few stress days, rho_stress is noisy. Report n and consider confidence intervals (folder 01 optional pages).

Look-ahead bias

Do not label stress using future knowledge of the pair correlation you are measuring.

09 / WORKED EXAMPLE, PART ONE

Calculate by hand: quiet subsample

Quiet regime (four made-up daily returns, decimals not percent labels):

Day	x	y
1	0.010	0.008
2	0.020	0.018
3	0.015	0.014
4	0.012	0.011

```
Means: x_bar = 0.01425, y_bar = 0.01275
Pearson steps give rho_quiet = about 0.9912
```

All pairs move up together in this toy quiet window, so r is very high. Real quiet weeks need not look this clean.

Calculate by hand: stress subsample

Stress regime (four made-up returns):

Day	x	y
1	-0.050	-0.030
2	0.040	-0.020
3	-0.030	0.025
4	0.035	-0.015

```
rho_stress = about -0.1895 in this toy list
```

Stress rho can differ sharply from quiet rho. That is the point. Do not merge the eight rows and pretend one number tells both stories.

11 / EFFECTIVE PRACTICE

Using regimes on FX majors

Compare dollar-block pairs in quiet vs high-vol weeks when your state rule is fixed and auditable. Store the state series alongside returns.

What not to do

Do not tune the stress threshold until rho_stress looks exciting. Do not average regime rhos for a headline.

Experiments not started; no live pair claims in this PDF.

Limits and software

Limits

Regimes are discrete; boundary days flip labels. Correlation within a thin stress slice is unstable. State rules are not causal models.

Python sketch

```
import numpy as np
# mask quiet rows, then np.corrcoef(x[mask], y[mask])[0,1]
```

Match Pearson definition to folder 01. Save n_quiet and n_stress in outputs.

13 / QUICK QUESTIONS

Check your understanding

Is rho_quiet a percent?

No. It is a correlation on the quiet subsample only.

Can I average rho_quiet and rho_stress?

Generally no as a substitute for the full-sample correlation or a properly pooled estimator.

Must states be defined in advance?

Yes for honest research. Do not pick stress days after seeing the pair correlation you want.

When experiments start, save the state indicator series next to returns so reviewers can reproduce each rho.

Sources and further reading

Is rho a percent? No. **Can I average regime rhos?** Not as a substitute for full-sample or pooled estimators. **Can I use future data to tag stress?** Not if you want an honest research rule.

Links checked 3 November 2026.

[1] Benediktsdottir, J., & Scotti, C. (2009). Exchange Rate Correlations in Quiet and Turbulent Times.
IMF working paper series; verify exact title and number on IMF site.
Open source

[2] Folder 01 Pearson Correlation guide.
Pearson steps used inside each regime.
Open source